

KAN performance on REAL dataset,

Using EPFL challenge 2016 dataset:

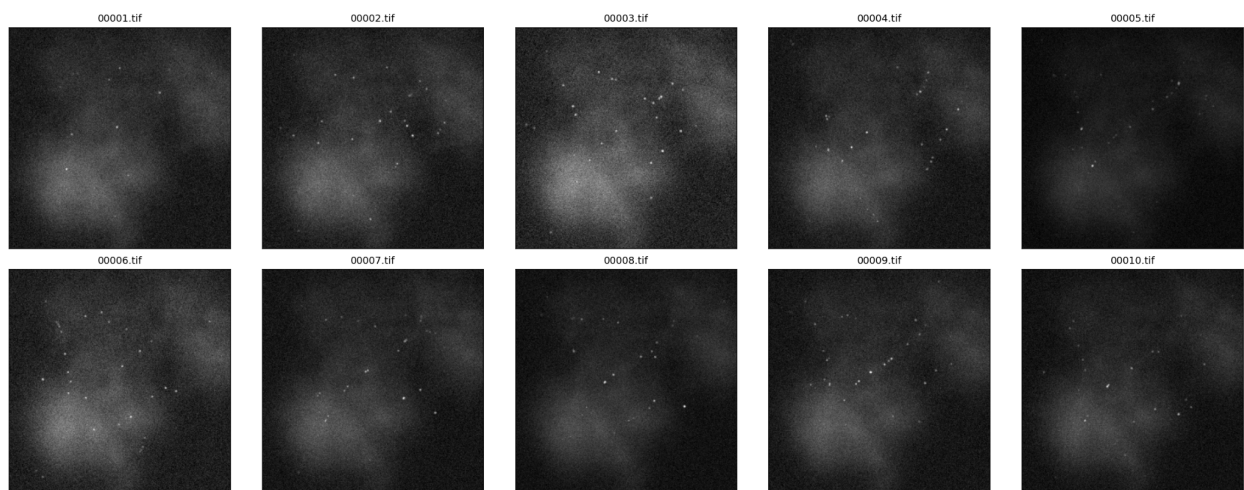
First I wanted to use their synthesized dataset for training the model. But the size of their training dataset(synthesized) is different with real datasets.

So I used my synthesized multi emitter dataset for training and validation and for test REAL data.

Training data in EPFL challenge website:

Pixels: 256*256 px

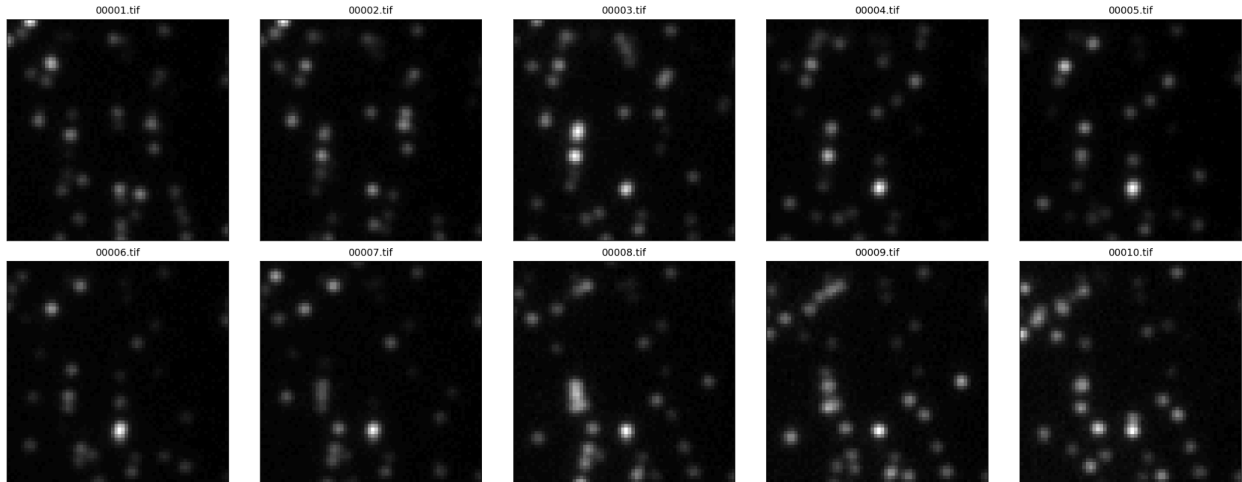
Size: 128nm



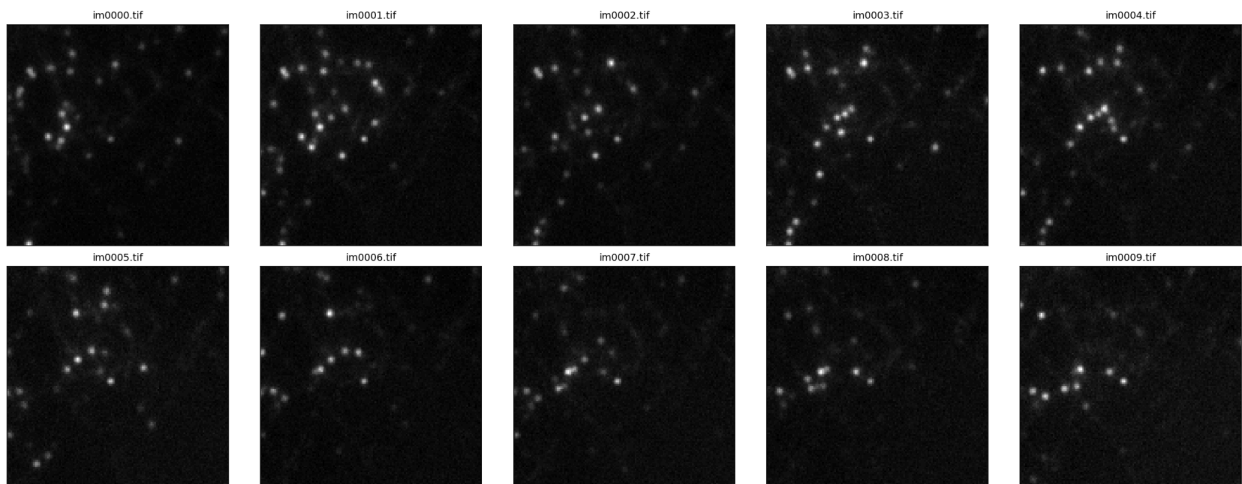
Test-Real data:

1. First dataset: low density
15000 frames

Real long sequence:



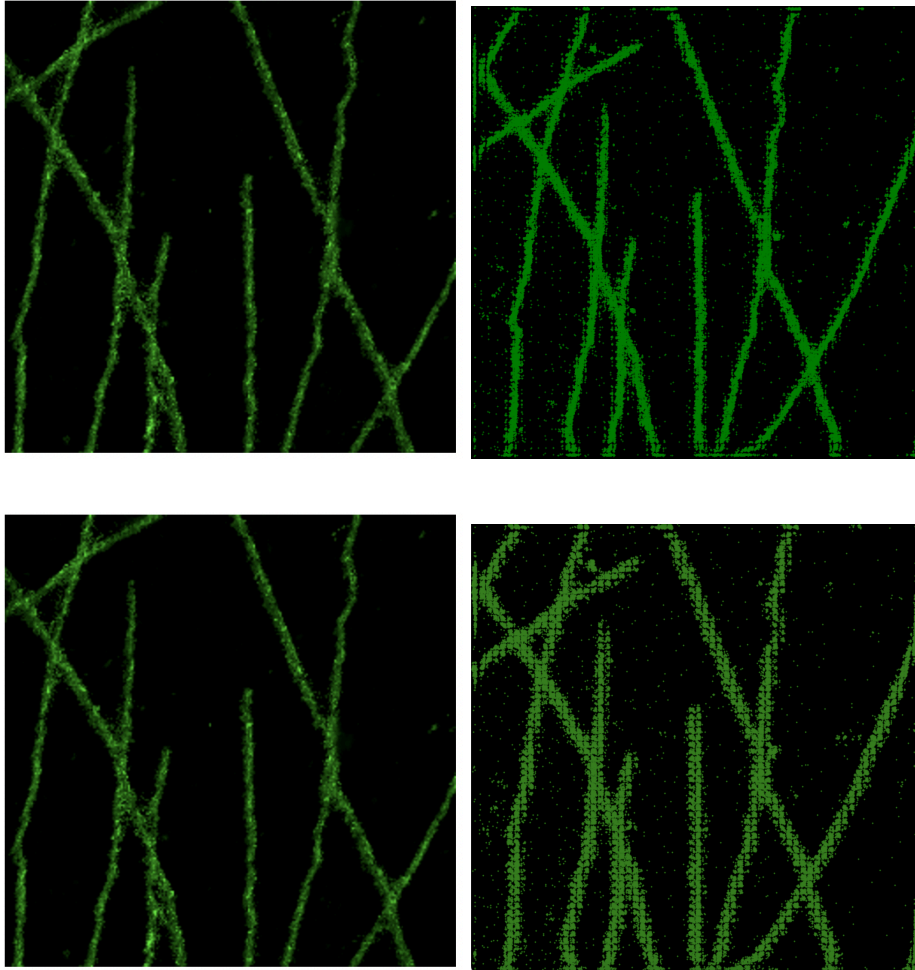
2. Second dataset: high density
9990 frames
Tubulin-AF647



Result:

Note: The ground truth does not exist for these dataset so the only way is to compare the picture from website

1. 4 hours



7 hours for training

3 hours for test

Training dataset:

8000 frame

12000 emitters

MD 2D GUASS

Model performance on validation: RMSE: 18 nm

